

WASP-12b

R-0332

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_231218 · created 2026-07-18T05:05:00+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460296.7754 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.0067
Transit depth (Rp/Rs)^2	2.4 +/- 0.21 [%]
Orbital Inclination (inc)	86.55 +/- 2.21
Airmass coefficient 1 (a1)	1.025 +/- 0.015
Airmass coefficient 2 (a2)	-0.022 +/- 0.015
Scatter in the residuals of the lightcurve fit is	1.46 %
Best Comparison Star	#1 - [206, 236]
Optimal Aperture	3.85
Optimal Annulus	12.87
Transit Duration (day)	0.1332 +/- 0.0024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.155 $\pm$ 0.0067 vs 0.1178 (deviation 5.55 σ)
Duration fit vs published	0.1332 d vs 0.125 d (deviation 3.42 σ)
Mid-time O-C	1.9 min
Depth SNR	23.1
Residual scatter	1.46 %

Light curve

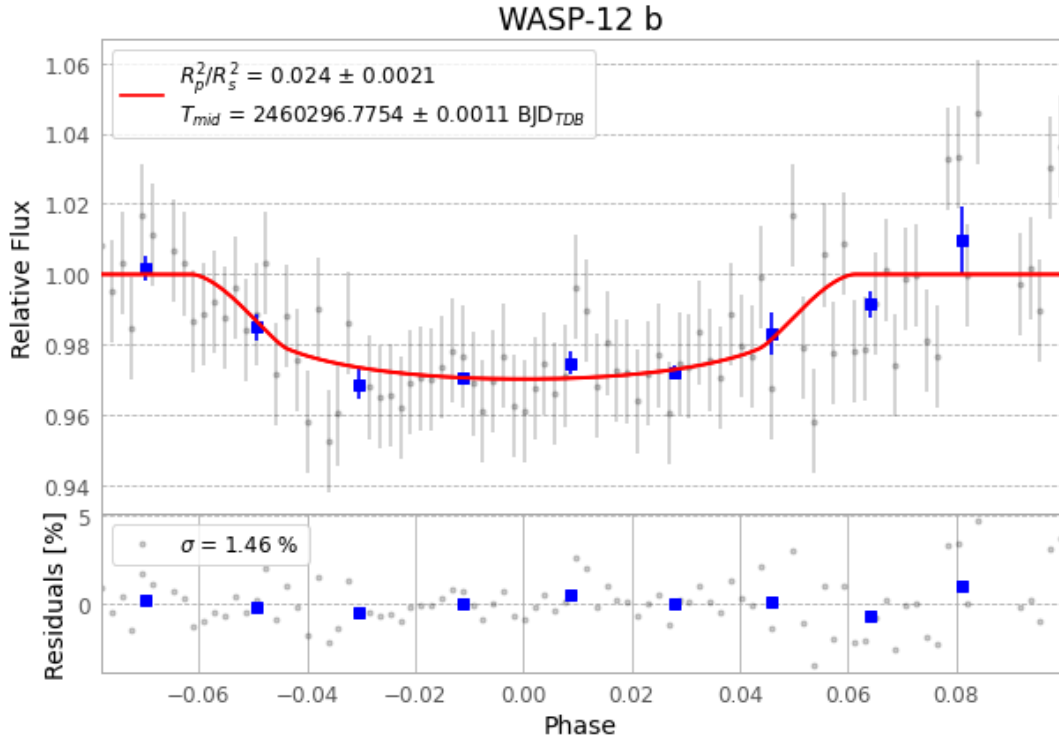


Figure 1 — FinalLightCurve_WASP-12 b_2023-12-17.png (SHA-256 d14f2e319c93d4e6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [319, 253], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5ebd4d15e4be0ce8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 600780d

Data provenance

94 FITS frames (62 MB), WASP-12231218042415.FITS → WASP-12231218090315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-231218.log (8e3a3c0a9038...), FinalLightCurve_WASP-12 b_2023-12-17.png (d14f2e319c93...), FinalLightCurve_WASP-12 b_2023-12-17.csv (23bf4b24a75f...)

Compute resources

Wall time 180.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 05:05Z.